

emPOWER NOVA

Work Completed & DNA Handoff

Prepared for the **SWARM** team
IoT Telemetry → Control (HMI) → AI OPERATOR

July 2026 · Shareable engineering brief
Companion: NOVA_DNA_source_for_SWARM.zip

1. What NOVA is

NOVA is emPOWER's controlled intelligence layer: natural-language questions become **catalog plans**, permission-filtered **skills**, certified **facts**, and **guarded narration**. It is not a free ChatGPT-with-database agent.

Product surfaces today include the ERP chat bubble, dedicated **NOVA Chat** (web + Android), Control Centre copilot (staff plane), and the **Reader** plane (document OCR → form draft). All chat planes share one server pipeline; Reader is a separate assistive plane that never posts ledger rows.

Architecture principles (NOVA DNA)

- **Plan** → **tools/skills** → **facts** → **narrate** — LLMs never pick tools freely or run SQL.
- **Rules-first** Search Engine; **NovaThink** only when plan confidence is low.
- **RBAC additive** — filter tools with `novaCanRunTool / can()`; never fail-open.
- **Read-only by default** — dual write guards; Safe Workflow may open forms with prefill but user must submit.
- **Clarify over guess** — ambiguous entities ask chips; bind by id.
- **Answer guards** — money, counts, identity, and period cannot drift after narration.

2. Engines inventory

Product vocabulary lists **12 engines** on the chat/reader planes. **NovaPlan** is the plan gate bridging understand → execute. **NovANALYSER** is the cross-module orchestration skill (meta-engine routing).

#	Engine	Purpose
1	Search Engine	Rules-first slot fill: family, entity, metric, period, depth, deny_write.
2	Intent & Lexicon	Synonyms and topics map to catalog tools when Search is not decisive.
3	NovaThink	Gated LLM slot polish only when plan confidence is low; catalog-validated.
4	Dialog / Sticky	Session dialogue state: bound entities, pending clarify, TTL, follow-ups.
5	Clarify	Ask-over-guess chips; bind by id (1 / code / label) — never re-fuzzy.
6	Entity Resolve	Permission-aware party / staff / project disambiguation.
7	Skills Catalog	Registered read-only skills + RBAC dispatch (Prisma under can()).
8	Packs	Multi-metric business chapters (month, collection, project command, ...).
9	Analysis Engine	Why is it here? Factor packs + ranked drivers for one domain.
10	Trend Engine	Who/what over time? Windowed series + rankings.
11	Present / Narrate	Deterministic format + guarded LLM polish + answer guards.
12	Reader Engine	OCR / vision to form prefill drafts — never posts ledger rows.
—	NovaPlan	Single plan gate: Search vs Lexicon/Think → tools, completeness, clarify.
—	NovANALYSER	Cross-module fan-out: correlate metrics, rank issues, narrate with evidence.

Core loop: `answerNovaQuery` → `DialogState` → Search / Think / Lexicon → `NovaPlan` → Clarify / Entity Resolve → RBAC → Skills (Packs / Analysis / Trend / NovANALYSER) → Present / Narrate.

3. Work completed (high-level)

Recent emPOWER deliveries that strengthen the foundation Swarm can reuse for AI OPERATOR:

- **Intelligence stack maturity** — Search, Lexicon, gated Think, DialogState, Clarify, Entity Resolve, Skills (~80+ catalog tools), Packs, Analysis, Trend, Present/Narrate, Reader.
- **NovANALYSER P0** — cross-module fan-out with correlation, ranking, and RBAC-filtered steps (business health / income health style intents).
- **Safe Workflow / FILL** — create-intent phrases open the correct form with safe prefill; never auto-submit (prefill ≠ commit).

- **Tenant / org isolation** — SaaS organization scope stamped on NOVA finance, ops, and HR skills; deny cross-org lookups.
- **NOVA Chat mobile** — dedicated Android client (BPG + SaaS), bearer client API, threads, clarify chips, reports, MFA; mic dictate + TTS polish.
- **Continued conversation** — sticky follow-ups (yes / ok / more), pending offers, session DialogState clear.
- **Control Centre NOVA** — floating copilot on staff plane with grounded cc . * tools, diagnostics/triage, kill-switch, repair propose as dry-run only with human confirm.
- **Commercial invariants** — catalog-only tools, dual write guards, immutable report snapshots, permission help when RBAC blocks.

4. How Swarm can reuse NOVA DNA

Swarm remains a **standalone** IoT product (telemetry SoT). emPOWER integrates via partner API where Manufacturing Hub needs physical truth. For Swarm's own AI OPERATOR, port the **orchestration pattern** and engines — do not embed ERP ledger skills.

Telemetry (now)	IoT dashboard: ingest, plants, alerts, health scores	Search + Lexicon for intent; Skills over local aggs; Present with digit/unit guards
Control (via HMI)	Operator HMI: setpoints, modes, safe commands with human confirm	Dialog + Clarify for ambiguous devices; Safe Workflow pattern (prefill ≠ commit); write-deny for unsafe acts
AI OPERATOR	Read-only insights → proposed actions with evidence → human gate	NovaPlan orchestration; NovaThink only low-conf; NovANALYSER-style fan-out across telemetry modules; RBAC allowlists

Suggested P0 for Swarm AI OPERATOR

Start **read-only**: natural-language telemetry insights using a NovANALYSER-style fan-out (intent → plan of metric skills → concurrent fetch → correlate → rank → narrate with unit/quality guards). No setpoint writes in P0. Keep a tool allowlist and plant/org RBAC on every step.

5. Suggested integration points

- **Study / port** — `src/lib/nova/`, `src/lib/novanalyser/`, `src/lib/ai/nova-plan.ts`, `nova-engine-routing.ts`, `nova-tool-permissions.ts`, `nova-think.ts`, DialogState, Search Engine.
- **Wrap as API (emPOWER side)** — when ERP users ask plant questions, NOVA skills call emPOWER's Swarm client over quality-flagged aggregates (soft-fail); never let NOVA hit Swarm DB.
- **Client pattern** — Android NOVA Chat + `/api/client/v1/nova/*` shows bearer + RBAC chat contracts Swarm can mirror for an Operator app.
- **RBAC** — copy the floor map pattern (tool → permissions any-of) plus handler `can ( )`; feature flags disable features, never open data.
- **HMI Control** — reuse Safe Workflow / Reader draft bridge ideas: propose → preview → human confirm; never silent write.

6. Contact & next steps

Mike's team (emPOWER)

Primary contact: *[Name / email — fill in]*

Secondary: *[Name / email — fill in]*

Suggested next steps

1. Open NOVA_DNA_source_for_SWARM.zip in Cursor with CURSOR_INSTRUCTIONS_FOR_SWARM.md.
2. Map Swarm telemetry domains → skill stubs (read-only).
3. Prototype AI OPERATOR P0: insights only, plant-scoped RBAC.
4. Align partner API contracts for emPOWER Manufacturing Hub pull (separate from AI OPERATOR).
5. Schedule joint design review: Control HMI command model + human-confirm gates.

Companion artifacts: source zip + Cursor instructions under output/nova-swarm-handoff/ and ~/Downloads/.